Test Case Suite

The following test suite spans critical functional, security, performance, and compliance workflows:

Test ID	Module	Test Scenario	Execution Steps & Inputs	Expected Result	Pass/Fail Criteria
TC-SEC-01	Auth	Biometric Authentication Bypass	1. Attempt login with manipulated JWT payload. 2. Replay captured biometric hash.	System rejects tampered token; biometric hash invalidation triggers account lock.	**Pass:** HTTP 401/403 returned, security alert logged.
TC-TXN-01	Payments	Valid Instant Money Transfer	1. Account A initiates transfer of $100 to Account B via UPI/IMPS. 2. Account A balance = $500.	Account A debited $100 ($400 remaining); Account B credited $100. Ledger entry created in a single ACID transaction.	**Pass:** Balances match double-entry ledger; total balance invariant preserved.

TC-TXN-02	Payments	Overdraft / Insufficient Balance	1. Account A initiates transfer of $1000. 2. Account A balance = $200.	Transaction declined instantly. No partial debits or database locks remain.	**Pass:** HTTP 422 returned with "Insufficient Funds"; balances unchanged.
TC-SEC-02	Security	SQL Injection on Search/Input	1. Inject ' OR '1'='1 into the payment remark and search bar fields.	Input sanitized at API gateway level; SQL parser receives parameterized inputs only.	**Pass:** Input rejected or treated strictly as literal string.
TC-KYC-01	Compliance	e-KYC Verification Process	1. Upload valid ID document & live selfie. 2. Trigger OCR and face matching API.	Face match confidence > 98%; ID verified against national registry.	**Pass:** Account status changes from PENDING to VERIFIED.

TC-CRD-01	Cards	Virtual Card Freeze Feature	1. Issue card freeze command from app. 2. Attempt POS transaction using card tokens.	Transaction rejected immediately by payment processor engine.	**Pass:** Auth request returned CARD_FROZEN.
TC-PERF-01	Platform	High Concurrency Load Test	1. Simulate 10,000 concurrent payment transactions/sec for 15 mins.	Average latency < 200ms; 0% transaction loss; database auto-scales without deadlock.	**Pass:** 99.9% requests successful within SLA bounds.

2. Software Maintenance & Operational Plan

Once deployed, the platform enters continuous operations under four primary maintenance domains:

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None
+-------------------------------------------------------------------
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|                          Software Maintenance Domains
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|
|
|   Preventive Maintenance                  Corrective Maintenance
|
```

```
|   - Daily automated vulnerability scans  - Zero-day exploit
patching          |
|   - Certificate renewals                 - P1 incident handling
(< 1 hr SLA)  |
|
|
|   Adaptive Maintenance                   Perfective Maintenance
|
|   - Central bank API updates             - Database query
optimization        |
|   - PCI-DSS spec updates                 - Dynamic caching
adjustments        |
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```

1. Corrective Maintenance (Bug Fixes & Patching)

- **P1 (Critical - Outage/Data Leak):** Resolution within **1 hour**. Dedicated war-room mobilization, instant hotfix deployment via continuous deployment pipeline.
- **P2 (High - Core Feature Impaired):** Resolution within **12 hours**. Fix included in emergency sprint patch release.
- **P3/P4 (Medium/Low - Cosmetic/Non-blocking):** Addressed in regular bi-weekly sprint cycles.

2. Adaptive Maintenance (Regulatory & Environment Changes)

- **Regulatory Compliance Updates:** Ongoing adjustments for new Central Bank mandate rules, updated tax withholding logic, or revised Open Banking API schemas.
- **Third-Party API Evolution:** Proactive migration when payment networks (e.g., Visa, Mastercard, SWIFT) release mandatory v2/v3 API specs.

3. Perfective Maintenance (Optimization & Tuning)

- **Database Performance Tuning:** Automated index optimization, table partitioning for transaction ledgers exceeding 100M rows, and Redis caching adjustments.
- **Latency Reduction:** API gateway refactoring and connection pooling to keep P99 transaction latency under 150ms.

4. Preventive Maintenance (Proactive Hardening)

- **Automated Security Audits:** Daily vulnerability scanning of base Docker images, dependencies (via Snyk/Dependabot), and IaC misconfigurations.

- **Certificate Renewal & Rotation:** Automated rotation of TLS certificates, API secret keys, and database credentials using HashiCorp Vault every 30 days.

3. SLA & Operational Monitoring Framework

Core Service Level Agreements (SLAs)

- **System Uptime Target:** 99.99% availability (less than 52.6 minutes of unscheduled downtime per year).
- **Payment Processing SLA:** 99.95% transaction success rate (excluding third-party clearinghouse outages).
- **Mean Time to Detect (MTTD):** < 2 minutes via automated synthetic monitoring.
- **Mean Time to Recover (MTTR):** < 15 minutes utilizing automated infrastructure rollbacks.

Observability Architecture

- **Logging:** Centralized structured JSON logging shipped to ELK Stack / Grafana Loki with zero storage of Plaintext PII or card numbers (PCI-DSS compliant masking).
- **Metrics:** Prometheus collection of microservice metrics (CPU, Memory, JVM Heap, DB Connection Pool saturation).
- **Tracing:** OpenTelemetry distributed tracing across all microservice calls to trace transaction propagation end-to-end.